LECTURE TWO : E- COMMERCE INFRASTRUCTURE

E-commerce Infrastructure: The Internet, Web, and Mobile Platform

Learning Objectives

- Discuss the origins of the Internet
- Identify the key technology concepts behind the Internet
- Describe the role of Internet programs and utility programs
- Explain the current structure of the Internet
- Understand the limitations of today's Internet
- Describe the potential capabilities of Internet II
- Understand how the World Wide Web works
- Describe how Internet and Web features and services support e-commerce

The Internet: Technology Background

- **Internet**
  - Interconnected network of thousands of networks and millions of computers
  - Links businesses, educational institutions, government agencies, and individuals
- **World Wide Web (Web)**
  - One of the Internet's most popular services
  - Provides access to billions, possibly trillions, of Web pages

The Internet: Technology Background

- **The World Wide Web (www, W3)** is an information system of interlinked hypertext documents that are accessed via the Internet and built on top of the Domain Name System.
- **DNS** is an abbreviation for Domain Name System, a system for naming computers and network services that is organized into a hierarchy of domains. DNS naming is used in TCP/IP networks, such as the Internet, to locate computers and services through user-friendly names.



The Evolution of the Internet 1961–Present

- **Innovation Phase, 1964–1974**
  - Creation of fundamental building blocks
    - What are these building blocks? Packet switching hardware, client server computing and communications protocol called TCP/IP
  - The initial purpose of Internet in the 1960s was to link large mainframe computers on different college campuses

The Evolution of the Internet 1961–Present Cont..

- **Institutionalization Phase, 1975–1995**
  - Large institutions provided funding and legitimization (legalization) for the invention called Internet e.g. Department of Defense and the National Science Foundation. Support from DoD ($ 1m) led to what was called ARPANET (Advanced Research Projects Agency Network)

The Evolution of the Internet 1961–Present Cont..

- **Commercialization Phase, 1995–present**
  - In 1986 NSF (National Science Foundation) took the responsibility to develop a civilian Internet then called NSFNET
  - From 1995 Private corporations take over, expand Internet backbone and local service to ordinary citizens

Development of the Internet Timeline

Figure 3.2, Page 121

FIGURE 3.2 DEVELOPMENT OF THE INTERNET TIMELINE

YEAR	EVENT	SIGNIFICANCE
INNOVATION PHASE 1961–1974		
1961	Leonard Kleinrock (MIT) publishes a paper on "packet switching" networks.	The concept of packet switching is born.
1961	Lawrence Roberts (MIT) connects a computer in Cambridge to a computer in California using a low-speed data line.	This is the first demonstration of a wide area network[a] using telephone circuits. It demonstrated that computers could retrieve data and run programs remotely (and that the phone system was too noisy and slow for the purpose).
1962	J.C.R. Licklider (MIT) writes memos calling for a "Galactic Network" of computers.	**The vision of a global computer network is born.**
1963	Licklider heads up Department of Defense (DOD) ARPA[b] network development.	This is the beginning of military interest and funding. ARPA becomes the largest funder of early Internet efforts.
1966	Lawrence Roberts convinces ARPA to fund development of ARPANET using packet switching.	The first effort begins to build a global packet-switched wide-area network.

[a]A "wide area network" is a network that connects computers and other devices over distances larger than 1,000 meters. Early mainframe computers had "local" networks connecting hundreds or thousands of dumb terminals usually in the same building (less than 1,000 meters apart) to centralized mainframe computers. These early networks essentially sent keyboard and display commands back and forth from the terminals to the mainframe computers. The new wide area networks proposed in the 1960s were much more powerful: These new networks promised to permit remote computers to transfer entire files, send e-mail messages, and execute programs on a local computer.

[b]ARPA refers to the Advanced Research Projects Agency of the United States Department of Defense. ARPA is essentially the research and development unit within the Department of Defense; it historically invests in promising new technologies that could have military significance.

(continued)

Development of the Internet Timeline (cont'd)

Figure 3.2, Page 122

FIGURE 3.2 DEVELOPMENT OF THE INTERNET TIMELINE (CONTINUED)

YEAR	EVENT	SIGNIFICANCE
1968	ARPA requests quotes from various companies to build packet switches.	The concept of packet switching moves toward physical reality.
1969	The first packet switches are installed at UCLA and Stanford by Bolt, Beranek and Newman (BBN), a defense contractor.	**The concept of packet switching is realized in hardware.**
1969	The first packet-switched message is sent from UCLA to Stanford.	The communications technology hardware of the Internet is born—the first "Inter Network." The questions now are: How could this new technology be used? And for what purpose?
1972	E-mail is invented by Ray Tomlinson of BBN. The first ARPANET application is demonstrated at the ICCC[c] conference. Roberts wrote the first e-mail utility program permitting listing, forwarding, and responding to e-mails.	The first "killer app" of the Internet is born.
1973	Bob Metcalfe (XeroxPark Labs) invents Ethernet and local area networks.	**Client/server computing is invented**. Ethernet permitted the development of local area networks and client/server computing in which thousands of fully functional desktop computers could be connected into a short distance (<1,000 meters) network to share files, run applications, and send messages. Although the Apple and IBM personal computers had not yet been invented, at XeroxPark Labs, the first powerful desktop computers were created in the late 1960s. Xerox's Ethernet StarNetwork connected these early desktop computers into a functioning office network that could share files, programs, and send messages.
1974	"Open architecture" networking and TCP/IP concepts are presented in a paper by Vint Cerf (Stanford) and Bob Kahn (BBN).	**TCP/IP invented.** The conceptual foundation for a single common communications protocol that could potentially connect any of thousands of disparate local area networks and computers, and a common addressing scheme for all computers connected to the network, are born. These developments made possible **"peer-to peer" "open" networking.** Prior to this, computers could only communicate if they shared a common proprietary network architecture, e.g., IBM's System Network Architecture. With TCP/IP, computers and networks could work together regardless of their local operating systems or network protocols.

[c]ICCC stands for International Computer Communication Conference, one of the first technical engineering groups devoted to communications among computers.

(continued)

Development of the Internet Timeline (cont'd)

Figure 3.2,

Page 123

FIGURE 3.2 DEVELOPMENT OF THE INTERNET TIMELINE (CONTINUED)

YEAR	EVENT	SIGNIFICANCE
INSTITUTIONAL PHASE 1980–1993		
1980	TCP/IP is officially adopted as the DOD standard communications protocol.	The single largest computing organization in the world adopts TCP/IP and packet-switched network technology.
1980	Personal computers are invented.	Altair, Apple, and IBM personal desktop computers are invented. These computers become the foundation for today's Internet, affording millions of people access to the Internet and the Web.
1983	ARPA creates a separate military network (MILNET) and ARPANET contains only civilian university traffic.	The idea of a "civilian" Internet is born.
1983	Telnet and File Transfer Protocol (FTP) services are deployed.	Telnet permits remote computers to link into a local computer and operate programs. FTP permits easy file transfers on the Internet. These services join e-mail as new "killer apps."
1984	Apple Computer releases the HyperCard program as part of its graphical user interface operating system called Macintosh.	The concept of "hyperlinked" documents and records that permit the user to jump from one page or record to another is commercially introduced.
1986	The National Science Foundation (NSF) adopts the Internet as its interuniversity network.	NSF begins a $200 million program to develop a university network. It requires all universities receiving NSF funds to make access available campus-wide.
1988	NSF encourages development of private long-haul backbone communication carriers to privatize the Internet.	Private firms such as PSI and UUNet form to handle commercial Internet traffic.
1989	Tim Berners-Lee of the physics lab CERN in Switzerland proposes a worldwide network of hyperlinked documents based on a common markup language called HTML—HyperText Markup Language.	**The concept of an Internet-supported service called the World Wide Web based on HTML pages is born.** The Web would be constructed from "pages" created in a common markup language, with "hyperlinks" that permitted easy access among the pages. The idea does not catch on rapidly and most Internet users rely on cumbersome FTP and Gopher protocols to find documents.
1990	NSF plans and assumes responsibility for a civilian Internet backbone and creates NSFNET.[d] ARPANET is decommissioned.	The concept of a "civilian" Internet open to all is realized through non-military funding by NSF.
1990	The Internet backbone grows.	By 1990, the backbone had grown from six nodes connected at 56 Kbps to 21 nodes connected at 45 Mbps. 50,000 networks were now connected on all continents, with 29,000 in the United States alone.
1993	The first graphical Web browser called Mosaic is invented by Mark Andreesen and others at the National Center for Supercomputing at the University of Illinois.	Mosaic makes it very easy for ordinary users to connect to HTML documents anywhere on the Web. The browser-enabled Web takes off.

[d]"Backbone" refers to the U.S. domestic trunk lines that carry the heavy traffic across the nation, from one metropolitan area to another. Universities are given responsibility for developing their own campus networks that must be connected to the national backbone.

(continued)

Development of the Internet Timeline (cont'd)

Figure 3.2, Page 124

FIGURE 3.2 DEVELOPMENT OF THE INTERNET TIMELINE (CONTINUED)

YEAR	EVENT	SIGNIFICANCE
1994	NSF report plans the development of an "information superhighway supporting universities, business and civilians."	Congress and the President's Office propose the creation of a national information superhighway to support research, educational, commercial, and private interests.
1994	Andreesen and Jim Clark form Netscape Corporation.	The first commercial Web browser—Netscape—becomes available.
1994	The first banner advertisements appear on Hotwired.com.	**The E-commerce I era begins.**
COMMERCIALIZATION PHASE 1995–PRESENT		
1995	NSF privatizes the backbone, and commercial carriers take over backbone operation.	The fully commercial civilian Internet is born. Major long-haul networks such as ATT, Sprint, GTE, UUNet, and MCI take over operation of the backbone. Network Solutions (a private firm) is given a monopoly to assign Internet addresses.
1996	Internet2 Consortium is formed.	Thirty-four government agencies, universities, and business firms plan the development of an Internet 100 to 1,000 times faster than the existing Internet I.
1998	The U.S. federal government encourages the founding of Internet Corporation for Assigning Numbers and Names (ICANN).	Governance over domain names and addresses passes to a private nonprofit international organization.
2002	U.S. Department of Commerce extends its contract with ICANN until September 2003.	Amid growing calls for more government intervention in the governance of the Internet, and growing criticism of ICANN, the private corporation is given new life as an interim measure.
2003	The Internet2 Abilene high-speed network achieves 8 Gbps speeds in a transcontinental test.	A major milestone toward the development of ultra-high-speed transcontinental networks several times faster than the existing backbone is achieved.

SOURCE: Based on Leiner, et. al., 2000.

The Internet: Key Technology Concepts

- **Internet defined as network that:**
  - Uses IP addressing
  - Supports TCP/IP
  - Provides services to users, in manner similar to telephone system
- **Three important concepts:**
  - Packet switching
  - TCP/IP communications protocol
  - Client/server computing

Packet Switching

- **Slices digital messages into packets**
- **Sends packets along different communication paths as they become available**
- **Reassembles packets once they arrive at destination**
- **Uses routers**
  - Special purpose computers that interconnect the computer networks that make up the Internet and route packets
  - **Routing algorithms** ensure packets take the best available path toward their destination
- **Less expensive in that it does not require dedicated circuit, not wasteful than circuit-switching since it makes full use of all available communication lines and capacity**

Packet Switching

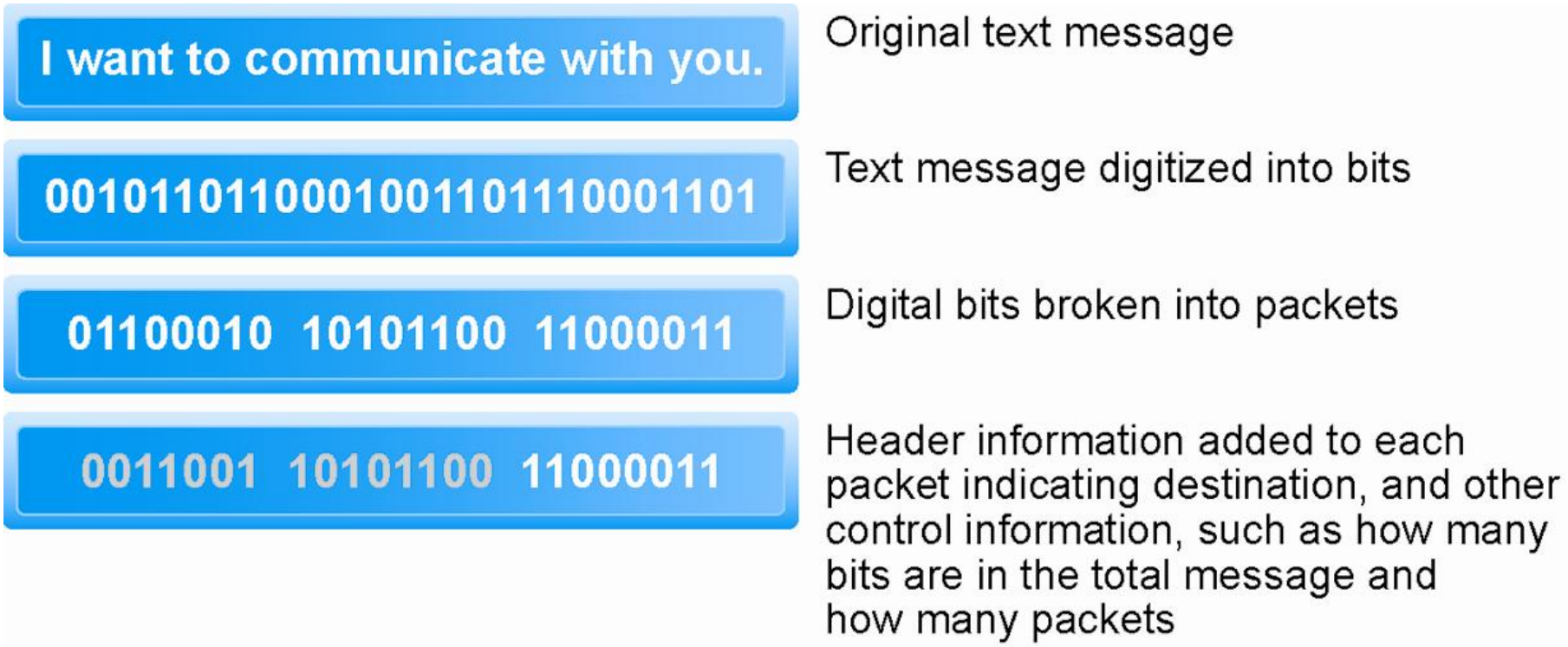


Figure 3.3, Page 117

TCP/IP

- **Transmission Control Protocol (TCP)**
  - Protocol: a set of rules for formatting, ordering, compressing and error-checking messages
  - Establishes connections among sending and receiving Web computers
  - Handles assembly of packets at point of transmission, and reassembly at receiving end
- **Internet Protocol (IP)**
  - Provides the Internet's addressing scheme
- **Four TCP/IP layers**
  - Network interface layer
  - Internet layer
  - Transport layer
  - Application layer

The TCP/IP Architecture and Protocol Suite

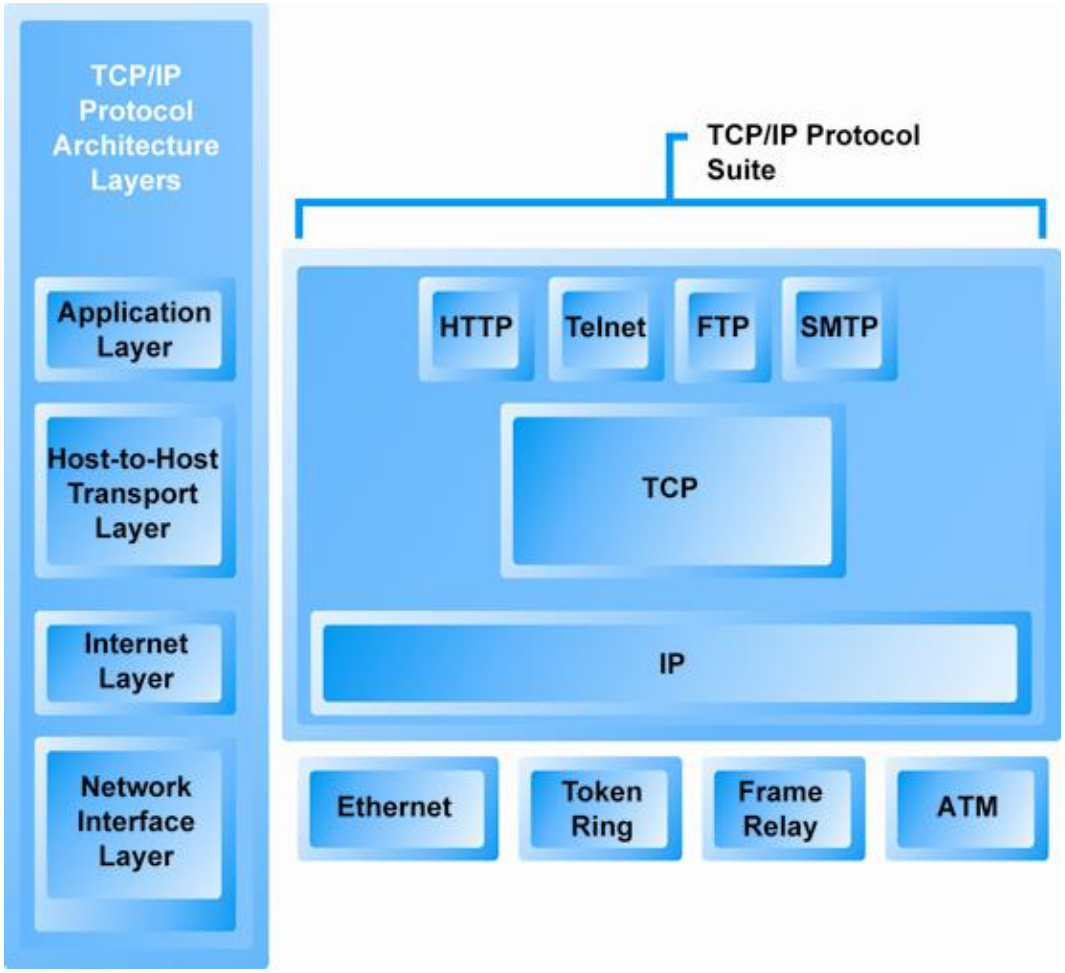


Figure 3.4, Page 119

Internet (IP) Addresses

- **IPv4**
  - 32-bit number
  - Four sets of numbers marked off by periods: 201.61.186.227
    - Class C address: Network identified by first three sets, computer identified by last set
- **IPv6**
  - 128-bit addresses, able to handle up to 1 quadrillion addresses (IPv4 can handle only 4 billion)

Routing Internet Messages: TCP/IP and Packet Switching

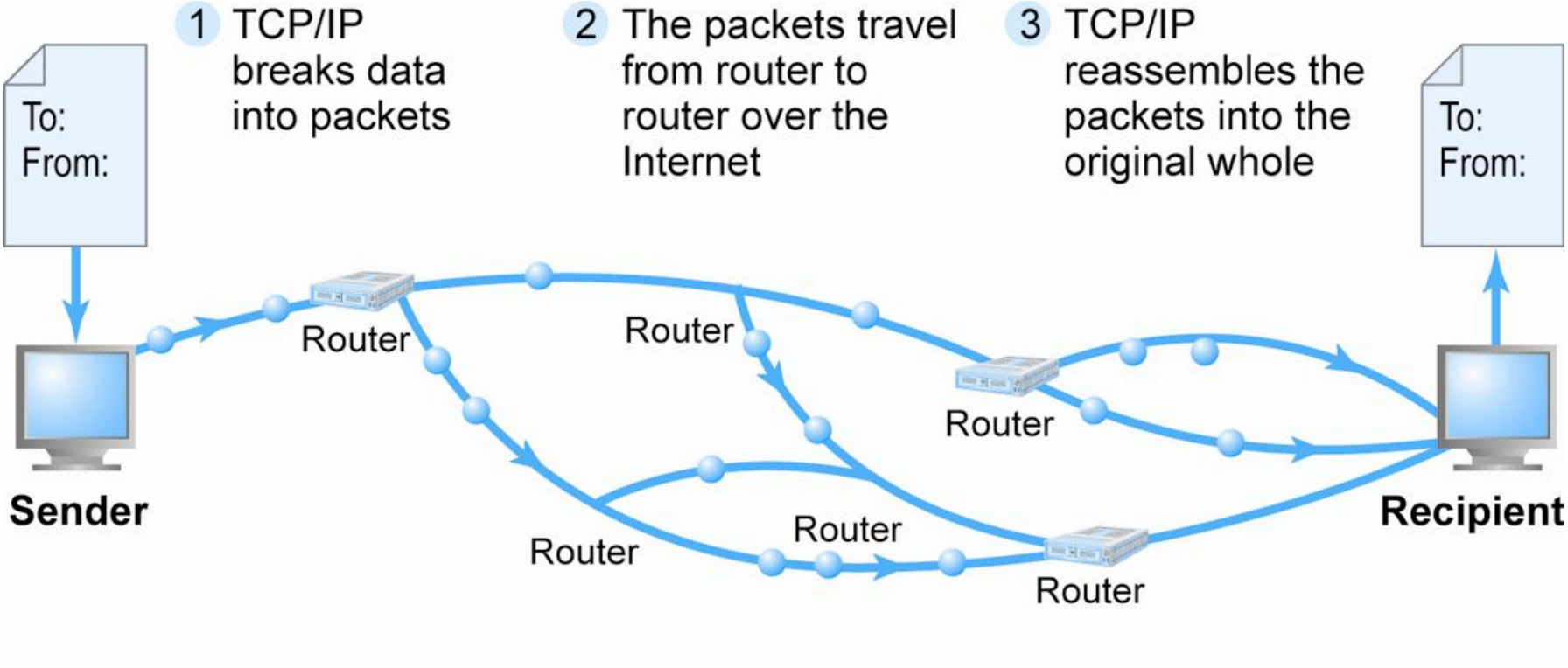


Figure 3.5, Page 120

Domain Names, DNS, and URLs

- **Domain name**
  - IP address expressed in natural language
- **Domain name system (DNS)**
  - Allows numeric IP addresses to be expressed in natural language
    - Example: cnet.com = 216.200.247.134
- **Uniform resource locator (URL)**
  - Address used by Web browser to identify location of content on the Web
  - For example: http://www.azimuth-interactive.com/flash_test
  - URL has four parts e.g name1.name2.name3.org

URLs


Uniform resource locator
Protocol used in the communication between the browser and the web server.
Indicates that a contact to a server is to be achieved.
Name of the domain where the web server belongs.
http://www.bbc.co.uk/news/
Colon separates the protocol from the other part of the web address.
Name of a server listening to messages using the http protocol.
Indicates a folder in the root folder.
Top-level Domain Name
http://www.library.cornell.edu/preservation/digital.html
Internet Protocol
URL Host Name
URL Path Name

Client/Server Computing

- **Powerful personal computers (clients) connected in network with one or more servers**
- **Servers perform common functions for the clients**
  - Storing files
  - Software applications
  - Access to printers, and so on

The New Client: The Mobile Platform

- **Presently, primary Internet access is through:**
  - Tablets
    - Supplementing PCs for mobile situations
  - Smartphones
    - Disruptive technology:
      - A disruptive technology is one that displaces an established technology and shakes up the industry or a ground-breaking product that creates a completely new industry.
        - Shift in processors, operating systems
    - ->33% of all cell phones

Cloud Computing

- **the practice of using a network of remote servers hosted on the Internet to *store, manage, and process data,* rather than a local server or a personal computer.**
- **Firms and individuals obtain computing power and software over Internet**
  - Example: Google Apps
- **Fastest growing form of computing**
- **Radically reduces costs of:**
  - Building and operating Web sites
  - Infrastructure, IT support
  - Hardware, software

Other Internet Protocols and Utility Programs

- **Internet protocols**
  - HTTP
  - E-mail: SMTP, POP3, IMAP
  - FTP, Telnet, SSL/TLS
- **Utility programs**
  - Ping
  - Tracert

The Internet Today

- **Internet growth has boomed without disruption because of:**
  - Client/server computing model
  - Hourglass, layered architecture
    - Network Technology Substrate
      - –composed of telecommunications networks and protocols.
    - Transport Services and Representation Standards-houses TCP/IP protocols
    - Middleware Services-a layer that ties the telecommunication networks to applications
    - Applications-consists of clients applications e.g www, e-mail etc.

The Hourglass Model of the Internet

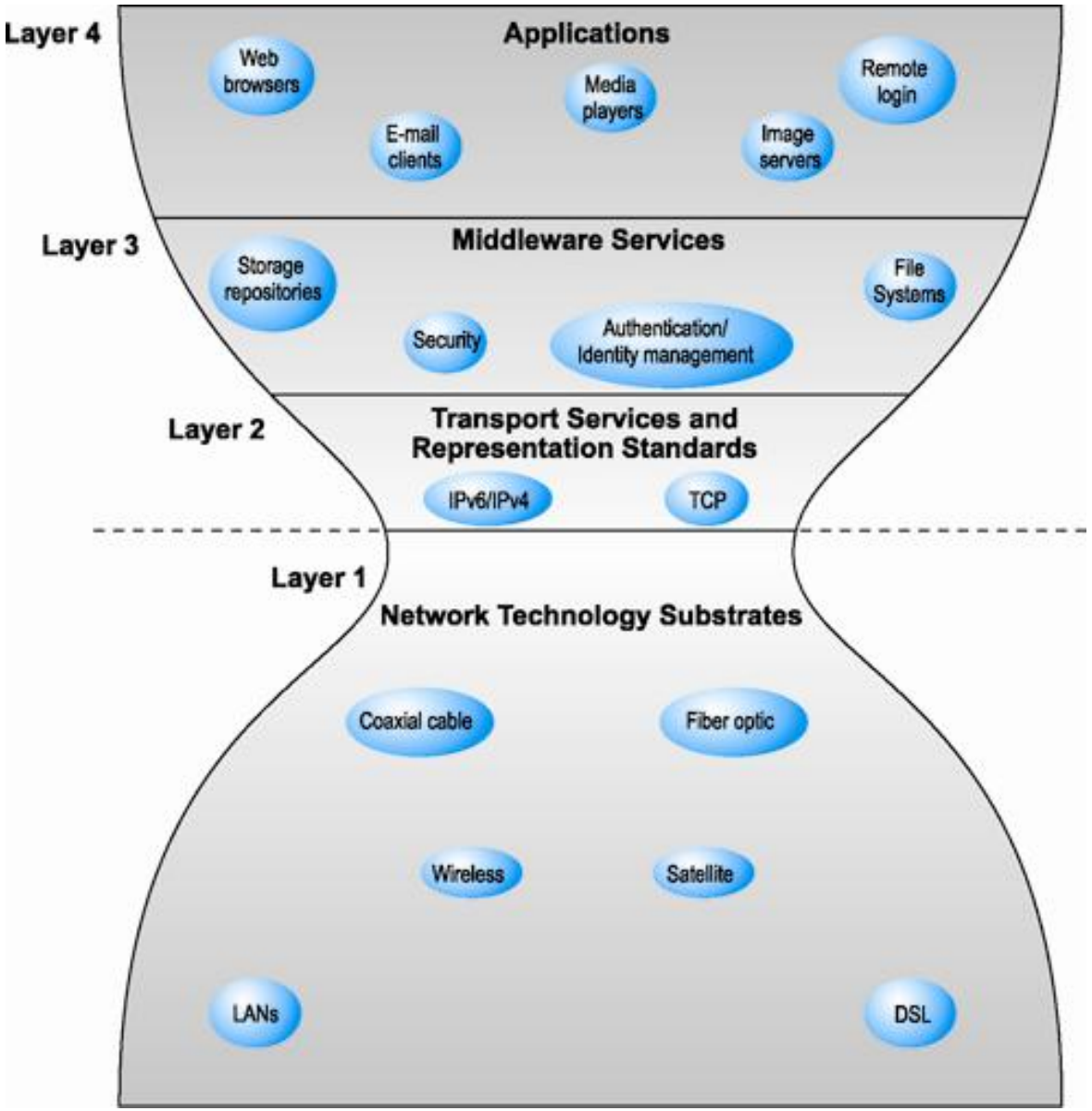


Figure 3.11, Page 128

Internet Network Architecture

- **Backbone**
  - High-bandwidth fiber-optic cable networks
  - Private networks owned by a variety of NSPs (Network Service Providers)
  - Bandwidth: 155 Mbps–2.5 Gbps----hihger
  - Built-in redundancy
- **IXPs (Internet Exchange Points)**
  - Hubs where backbones intersect with regional and local networks, and backbone owners connect with one another
- **CANs (Campus Area Networks)**
  - LANs operating within a single organization that leases Internet access directly from regional or national carrier

Internet Network Architecture

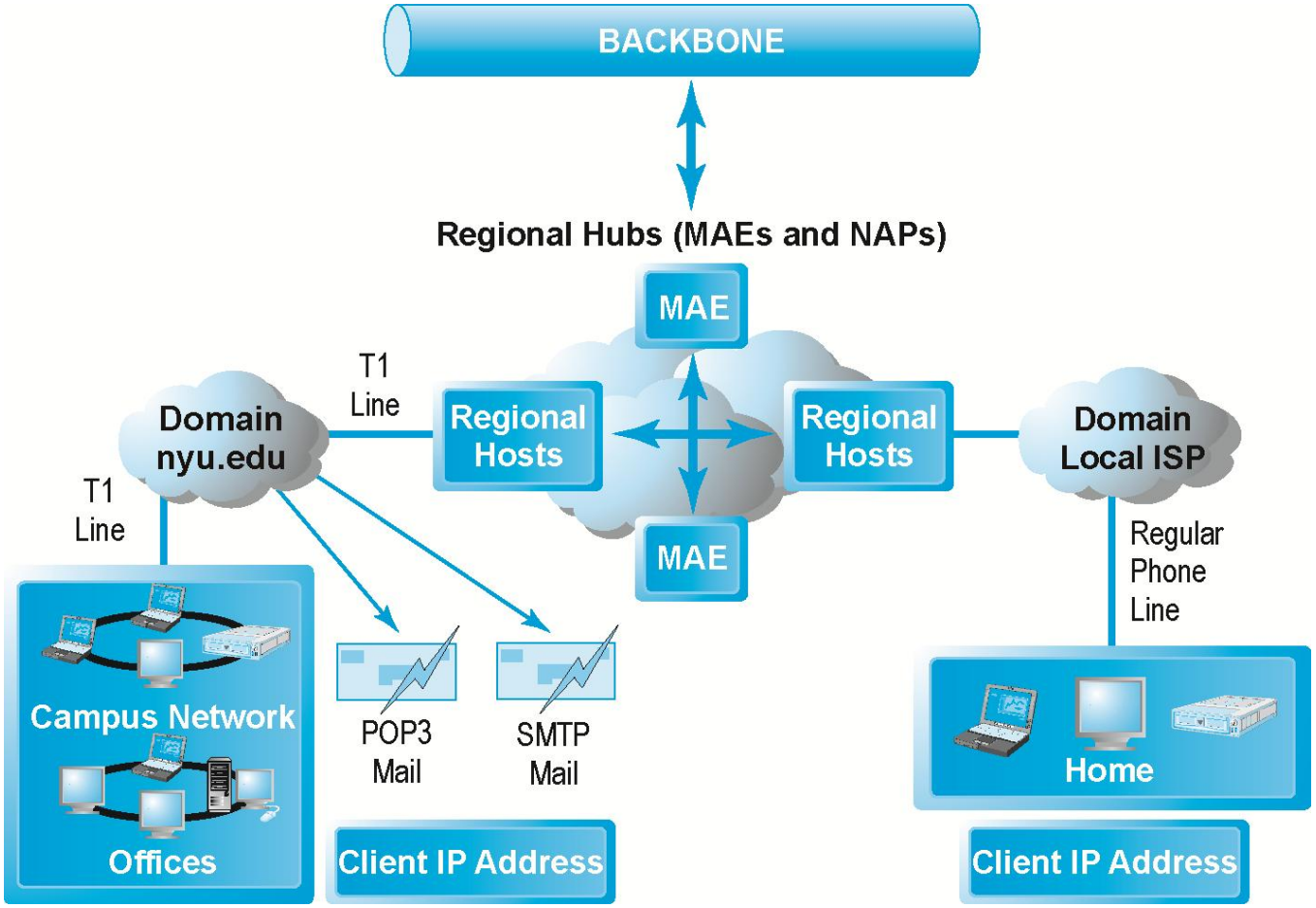


Figure 3.12, Page 129

Internet Network Architecture

MAEs(for Metropolitan Area Exchange point) or NAPs(Network Access points) are Internet Exchange points, they can be used interchangeably.

- MAE- Stands for Metropolitan Area Exchange point and NAP stands for Network Access Point

Internet Service Providers (ISPs)

- **Provide lowest level of service to individuals, small businesses, some institutions**
- **Types of service**
  - Narrowband (dial-up)
  - Broadband
    - Digital Subscriber Line (DSL)
    - Cable modem
    - T1 and T3
    - Satellite

Broadband Service Choices

- **Digital Subscriber Line (DSL): Telephone technology delivers high-speed access through ordinary telephone lines; speeds from about 385 Kbps to 1.5 Mbps**
- **Cable modem: Cable television technology allows digital access to Internet on top of analog video cable line; speeds from about 500 Kbps to 2 Mbps**
- **T1 and T3: International telephone standards for digital communication that offer guaranteed delivery rates**
  - T1: 1.54 Mbps
  - T3: 43 Mbps
- **Satellite: high-speed downloads (256 Kbps to 1 Mbps), but no upload available**

Intranets

- **Intranet**
  - TCP/IP network located within a single organization for communications and processing
  - Used by private and government organizations for internal networks
  - All Internet applications can be used in private intranets

Who Governs the Internet?

- **Organizations that influence the Internet and monitor its operations include:**
  - Internet Corporation for Assigned Names and Numbers (ICANN)
  - Internet Assigned Numbers Authority (IANA)
  - Internet Engineering Task Force (IETF)
  - Internet Research Task Force (IRTF)
  - Internet Engineering Steering Group (IESG)
  - Internet Architecture Board (IAB)
  - Internet Society (ISOC)
  - Internet Governance Forum (IGF)
  - World Wide Web Consortium (W3C)
  - Internet Network Operators Groups (NOGs)

Insight on Society: Class Discussion

Government Regulation and Surveillance of the Internet

- How is it possible for any government to "control" or censor the Web?
- Does the Kenya, Chinese government, or the U.S. government, have the right to censor content on the Web?
- What would happen to e-commerce if the existing Web split into a different Web for each country?

Limitations of the Current Internet

- **Bandwidth limitations**
  - Slow peak-hour service
- **Quality of service limitations**
  - Latency:
    - an expression of how much time it takes for a packet of data to get from one designated point to another
- **Network architecture limitations**
  - Identical requests are processed individually
- **Wired Internet**
  - Copper and expensive fiber-optic cables

The Internet 2 Project

- **Consortium of 350+ institutions collaborating to facilitate revolutionary Internet technologies**
- **Primary goals:**
  - Create leading-edge very-high speed network for national research community
  - Enable revolutionary Internet applications
  - Distributed and collaborative computing environments for sciences, health, arts, and humanities initiatives

The First Mile and the Last Mile

- **GENI Initiative Internet (Next Generation)**
  - Proposed by NSF to develop new core functionality for Internet
- **Most significant private initiatives**
  - Fiber optic trunk-line bandwidth
    - First mile
  - Wireless Internet services
    - Last mile

Fiber Optics and the Bandwidth Explosion in the First Mile

- **"First mile": Backbone Internet services that carry bulk traffic over long distances**
- **Fiber-optic cable: hundreds of glass strands that use light to transmit data**
  - Faster speeds and greater bandwidth
  - Thinner, lighter cables
  - Less interference
  - Better data security
- **Substantial investments in fiber optic by telecommunications firms in last decade**
  - Enable integrated phone, broadband access, video services

The Last Mile: Mobile Internet Access

- **"Last mile": From Internet backbone to user's computer, smartphone, and so on**
- **Two different basic types of wireless Internet access:**
  - Telephone-based (mobile phones, smartphones)
  - Wireless local area network (WLAN)-based

Wireless Internet Access Network Technologies

- **Wi-Fi**
  - High-speed, fixed broadband wireless LAN (WLAN)
  - Wireless access point ("hot spots")
  - Limited range but inexpensive
- **WiMax**
  - High-speed, medium range broadband wireless metropolitan area network
- **Bluetooth**
  - Personal connectivity between devices and to Internet
  - Low-speed, short range connection

Wi-Fi Networks

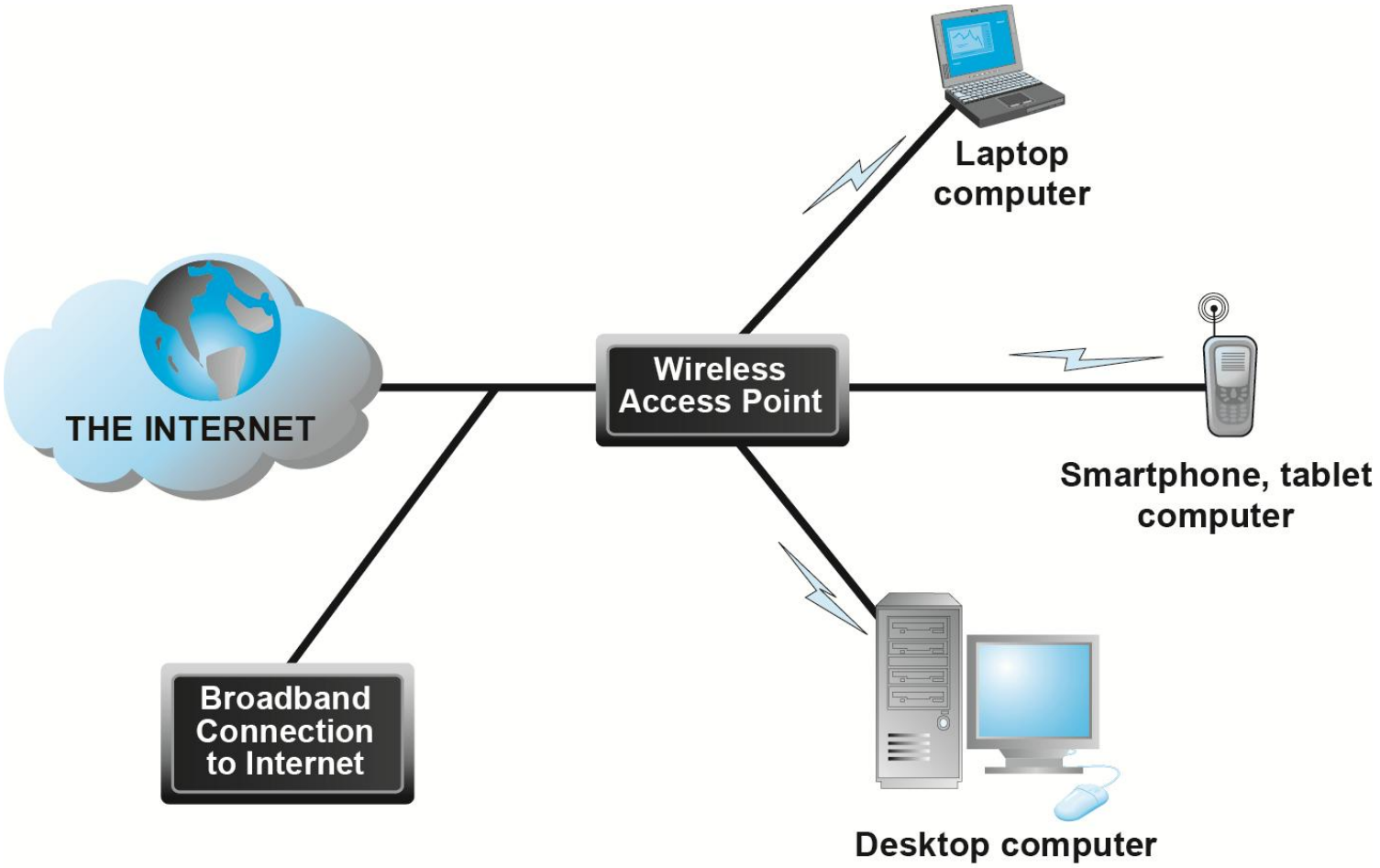


Figure 3.15, Page 145

The Future Internet

- **Latency solutions**
  - diffserv (differentiated quality of service)
- **Guaranteed service levels and lower error rates**
  - Ability to purchase the right to move data through network at guaranteed speed in return for higher fee
- **Declining costs**
- **The Internet of Things (IoT)**
  - Objects connected via sensors/RFID to the Internet
  - Spearheaded by EU and China

The Web

- **1989–1991: Web invented**
  - Tim Berners-Lee at CERN
  - HTML, HTTP, Web server, Web browser
- **1993: Mosaic Web browser GUI**
  - Andreessen and others at NCSA(National Centre for Supercomputing Applications)
  - Runs on Windows, Macintosh, or Unix
- **1994: Netscape Navigator, first commercial Web browser**
  - Andreessen, Jim Clark
- **1995: Microsoft Internet Explorer –Mosaic turned out to be IE.**

Hypertext

- **Text formatted with embedded links**
  - Links connect documents to one another, and to other objects such as sound, video, or animation files
- **Uses Hypertext Transfer Protocol (HTTP) and URLs to locate resources on the Web**
  - Example URL: http://megacorp.com/content/features/082602.html

Markup Languages

- **Hypertext Markup Language (HTML)**
  - Fixed set of pre-defined markup "tags" used to format text
  - Controls look and feel of Web pages
  - HTML5 the newest version
- **eXtensible Markup Language (XML)**
  - Designed to describe data and information
  - Tags used are defined by user

Is HTML5 Ready for Primetime?

- What features of HTML5 are changing the way Web sites are built?
- Is HTML5 a disruptive technology, and if so, for whom?
- Are there any disadvantages in Web sites and mobile apps moving to an HTML5 platform?

Web Servers and Web Clients

- **Web server software**
  - Enables a computer to deliver Web pages to clients on a network that request this service by sending an HTTP request
    - Apache, Microsoft IIS (Internet Information Server)
  - Basic capabilities: Security services, FTP, search engine, data capture
- **Web server**
  - May refer to either Web server software or physical server
  - Specialized servers: Database servers, ad servers, and so on
- **Web client**
  - Any computing device attached to the Internet that is capable of making HTTP requests and displaying HTML pages

Web Browsers

- **Primary purpose of web browsers is to display Web pages**
  - chrome—>70%
  - Mozilla Firefox—15%
  - Internet Explorer—x% of market
    - Open source
  - Other browsers
    - Google Chrome—>70%
    - Apple's Safari—5%
    - Opera
    - Maxthon
    - Sea monkey
    - Safari
    - chromium

2017	Chrome	IE/Edge	Firefox	Safari	Opera
January	73.7 %	4.9 %	15.4 %	3.6 %	1.0 %

2016	Chrome	IE/Edge	Firefox	Safari	Opera
December	73.7 %	4.8 %	15.5 %	3.5 %	1.1 %
November	73.8 %	5.2 %	15.3 %	3.5 %	1.1 %
October	73.0 %	5.2 %	15.7 %	3.6 %	1.1 %
September	72.5 %	5.3 %	16.3 %	3.5 %	1.0 %
August	72.4 %	5.2 %	16.8 %	3.2 %	1.1 %
July	71.9 %	5.2 %	17.1 %	3.2 %	1.1 %
June	71.7 %	5.6 %	17.0 %	3.3 %	1.1 %
May	71.4 %	5.7 %	16.9 %	3.6 %	1.2 %
April	70.4 %	5.8 %	17.5 %	3.7 %	1.3 %
March	69.9 %	6.1 %	17.8 %	3.6 %	1.3 %
February	69.0 %	6.2 %	18.6 %	3.7 %	1.3 %
January	68.4 %	6.2 %	18.8 %	3.7 %	1.4 %

The Internet and Web: Features

- **Features on which the foundations of e-commerce are built:**
  - E-mail
  - Instant messaging
  - Search engines
  - Online forums and chat
  - Streaming media
  - Cookies

E-mail

- Most used application of the Internet
- Uses series of protocols for transferring messages with text and attachments from one Internet user to another

Instant Messaging

- Displays words typed on a computer almost instantly, and recipients can respond immediately in the same way

Search Engines

- **Identify Web pages that match queries based on one or more techniques**
  - Keyword indexes, page ranking
- **Also serve as:**
  - Shopping tools
  - Advertising vehicles (search engine marketing)
  - Tool within e-commerce sites
- **Outside of e-mail, most commonly used Internet activity (Searching)**

How Google Works

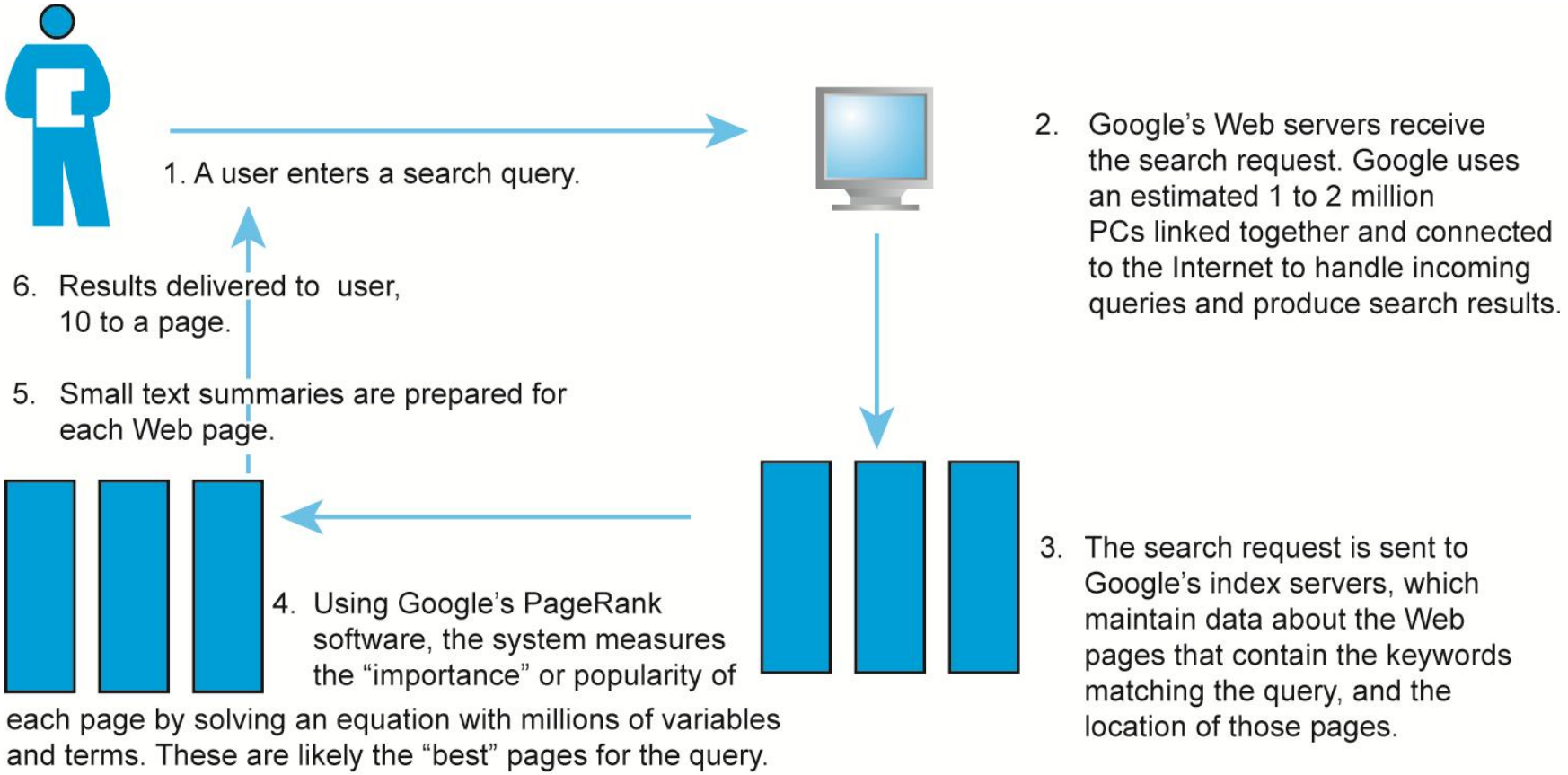


Figure 3.20, Page 161

Online Forums and Chat

Online forum

- Also known as a message board, bulletin board, discussion board, discussion group, board, or forum
  - *Web application that enables Internet users to communicate with one another, although not in real time*
- Members visit online forum to check for new posts

Online chat

- Similar to IM (Instant Messaging), but for multiple users
- Typically, users log into chat room

Streaming Media

- Enables music, video, and other large files to be sent to users in chunks so that when received and played, file comes through uninterrupted
- Allows users to begin playing media files before file is fully downloaded

Cookies

- Small text files deposited by Web site on user's computer to store information about user, accessed when user next visits Web site
- Can help personalize Web site experience
- Can pose privacy threat

Web 2.0 Features and Services

- **Online Social Networks**
  - Services that support communication among networks of friends, peers
- **Blogs**
  - Personal Web page of chronological entries
- **Really Simple Syndication (RSS)**
  - Program that allows users to have digital content automatically sent to their computers over the Internet

Web 2.0 Features and Services

- **Podcasting**
  - Audio presentation stored as an audio file and available for download from Web
- **Wikis**
  - Allows user to easily add and edit content on Web page
- **Music and video services**
  - Online video viewing
  - Digital video on demand

Web 2.0 Features and Services

- **Internet telephony (VoIP)**
  - Voice over Internet Protocol (VoIP) uses Internet to transmit voice communication
- **Video conferencing, video chatting, and telepresence**
- **Online software and Web services**
  - Web apps, widgets, and gadgets

Intelligent Personal Assistants

- **Software that interacts with the user through voice commands**
- **Features**
  - Natural language; conversational interface
  - Situational awareness
  - Interpret voice commands to interact with various Web services
- **Examples:** Siri, Google Now

Mobile Apps

- **Use of mobile apps has exploded**
  - More than 60% of online shoppers are mobile shoppers as well
- **Increased use/purchasing from tablets**
- **Platforms**
  - iPhone/iPad (iOS), Android, Blackberry
- **App marketplaces**
  - Google Play, Apple's App Store, RIM's App World, Windows Phone Marketplace

Apps for Everything: The App Ecosystem

- What are apps and why are they so popular?
- Do you use any apps regularly? Which ones, and what are their functions?
- What are the benefits of apps? The disadvantages?
- Are there any benefits/disadvantages to the proprietary nature of the Apple platform?

THE END OF LECTURE